Bence Visztra

Mechatronics & Power Electronics Engineer

Denmark | +45 91 67 75 58 | bence@billetraceworks.eu | Project Portfolio (Github)

Available for Full-Time Employment Immediately

Education

MSc in Advanced Power Electronics (Part-time) 2025 – 2027
Focus: Inverter topologies, motor drives, digital control systems, V&V project reports Aalborg University Esbjerg

BSc in Applied Industrial Electronics 2022 – 01/2026
Focus: Embedded hardware, sensors, analog/digital circuit design, V&V project reports Aalborg University Esbjerg

Professional Experience

Founder, Engineer, & Mechanical Fabricator 10/2025 – Present
Billet Raceworks Denmark

- In-house design and production of CNC milled custom high-stress chassis components, suspension linkages, and gearboxes in 6082-T6 and 7075-T6 aluminum.
- Handling assembly and BOM, dealing with suppliers for sourcing sheet metal fabrication work. Strong DFMA skills and experience in GD&T.
- Program multi-setup 3-axis CAM toolpaths (Fusion 360), design custom soft jaws and modular fixtures, holding ± 0.02 mm tolerances for press-fit bearings and alignment bores.
- Perform static FEA to optimize part mass while ensuring fatigue resistance under harsh dynamic road impacts.

E-Bike & Battery Systems Technician (Part-time) 10/2022 – 10/2025
Efuture ApS Denmark

- Diagnosed and repaired high-capacity Li-ion battery packs, brushless motors, and motor drivers.
- Designed and built custom battery packs using spot welding and BMS integration; operated CO2 laser cutter for custom parts.

Highlighted Engineering Projects

40 kW Custom Racing EV Platform Drivetrain & Structural Integration

- Engineered a race-winning 40 kW peak 4WD aluminum electric skateboard racing chassis and dual-stage reduction gearbox, designed from raw stock to final assembly.
- Built an ~IP65 sealed structural enclosure with integrated heatsinks, cable glands, and modeled wiring harnesses for ESC arrays and battery wiring.
- Conducted fatigue life calculations, fastener and bearing sizing calculations, and static loading FEA to ensure a lightweight yet strong chassis.
- Implementation of IMU sensors and steering angle measurement with hall effect sensors is currently in progress for a planned torque vectoring traction control system.

STM32-Based FOC Motor Controller (ESC) Power Electronics & Digital Control

- Implemented discrete Field-Oriented Control (FOC) on STM32 with Space Vector PWM (SVPWM), Park/Clarke transforms, and low-latency ADC sampling.
- Designed a compact 3-phase MOSFET inverter PCB with low-inductance DC-link busbars, isolated gate drivers, and inline shunt current sensing.
- Tuned current and velocity PI loops under variable dynamic load conditions.

Battery Health and Impedance Analyzer Embedded Diagnostics

- Developed an inverter board to perform Electrochemical Impedance Spectroscopy (EIS) routine measuring AC internal cell resistance to establish a battery equivalent circuit model.

Racing Line Optimizer and GPS Log Comparator Tool Data Analytics

- Developed a Matlab algorithm to find the optimal trajectory for specified vehicle parameters within track limits.

Technical Skills

CAD / CAM & Mechanics: Fusion 360, DFMA, GD&T, 2D production drawings, FEA (Fusion and Ansys), bearing & fastener sizing, enclosure sealing (IP ratings), BOM, dealing with suppliers for sheet metal work.
Workshop & Fabrication: 3-axis CNC milling, CAM toolpaths, fixture making, CO2 laser cutting, spot welding, MIG welding, 3D printing, manual assembly.
Electronics & Power: Schematic & PCB design (EasyEDA Pro), LTspice, 3-phase inverters, power conversion, motor drives, battery systems, wiring design, conductor sizing, bench instrumentation, BOM, ordering.
Embedded & Control: C, STM32 (HAL), ESP32, CAN bus, UART, SPI, I2C, discrete control, PID, MATLAB / Simulink.